

[illegible]
$$Y_1 = \binom{4}{\kappa_1} + \binom{4}{\kappa_2} \&\& \text{Det}(1^+) = -\frac{1}{2} k^2 (\binom{4}{\kappa_1} + \binom{4}{\kappa_2}) \&\& \text{Det}(2^+) = \frac{3}{2} k^2 (\binom{4}{\kappa_1} + \binom{4}{\kappa_2})$$
$$\begin{aligned} & \frac{4}{3} \binom{(4)}{K_1} \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^X \mathcal{K}^\alpha_\alpha{}^\beta + \binom{(4)}{K_2} \partial_X \mathcal{K}^\delta_{\beta\delta} \partial^X \mathcal{K}^\alpha_\alpha{}^\beta - \frac{4}{3} \binom{(4)}{K_1} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta - \\ & \frac{4}{3} \binom{(4)}{K_2} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta + \frac{2}{3} \binom{(4)}{K_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta + \frac{2}{3} \binom{(4)}{K_2} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta - \\ & \frac{5}{3} \binom{(4)}{K_1} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - \frac{5}{3} \binom{(4)}{K_2} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - \frac{1}{3} \binom{(4)}{K_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - \frac{1}{3} \binom{(4)}{K_2} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} \end{aligned}$$


A diagram showing two vertices connected by a horizontal line. Above the line, a green arrow points to the right, labeled k^μ . Each vertex has two external lines extending outwards, one upwards and one downwards.